

Question

Write a Java program to demonstrate the relationship **C1 implements I1**. Use constructors, method overriding and method overloading to calculate the area of a circle.

Steps

1. Create an interface **Shape**.
2. Declare a **calculate()** method in the interface.
3. Create a class **Circle**.
4. Make **Circle** implement **Shape**.
5. Declare radius in the class.
6. Create a constructor to initialize radius.
7. Implement and override the **calculate()** method.
8. Calculate the area of the circle.
9. Overload the **calculate()** method with different parameters.
10. Create a **Circle** object in **main()**.
11. Call the methods.
12. Display the area.

Algorithm

START

1. Create interface Shape.
2. Declare calculate() method.
3. Create class Circle implementing Shape.
4. Declare radius R.
5. Initialize R using constructor.
6. Implement calculate().
7. Calculate Area = $\pi \times R \times R$.

8. Overload calculate() with different parameters.
9. Create Circle object in main().
10. Call the calculate() methods.
11. Display the area.

STOP

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment2> c:; cd 'c:\Users\
25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp
e311af8ea878763765323f0f\redhat.java\jdt_ws\Assignment2_4e0b2320\bin
Area of Circle: 78.5
Area: 78.5
Area: 78.53975
PS C:\Users\venik\OneDrive\Assignmets\Assignment2>
```